Logistic Regression and Classification

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Logistic Regression & Classification Practice Exercises**

For the matrix $A = (1, 2, 0, 1; 2, 4, 1, 4; 3, 6, 1, 5)$, find bases for $\text{Row}(A)$, $\text{Col}(A)$, and $\text{Null}(A)$. Verify the Rank–Nullity Theorem.

Step 1: Row reduction to RREF: $(1, 2, 0, 1; 2, 4, 1, 4; 3, 6, 1, 5) \to (1, 2, 0, 1; 0, 0, 1, 2; 0, 0, 1, 2) \to (1, 2, 0, 1; 0, 0, 1, 2; 0, 0, 0, 0)$. Step 2: Subspace Bases:Basis for $\text{Row}(A)$: Non-zero rows of RREF: $\{(1, 2, 0, 1), (0, 0, 1, 2)\}$.Basis for $\text{Col}(A)$: Pivot columns of original matrix A (columns 1 and 3): $\{(1; 2; 3), (0; 1; 1)\}$.Basis for $\text{Null}(A)$: Free variables.

2 **[Foundational]** • **Target: Logistic Regression & Classification Practice Exercises**

A random sample of 200 students is classified by Grade Level (9th vs 12th) and Smartphone Operating System (iOS vs Android). Observed counts: 9th Grade: 70 iOS, 30 Android (100 total); 12th Grade: 50 iOS, 50 Android (100 total). (a) Calculate the expected count of 9th graders with iOS. (b) Compute the Chi-square test statistic χ^2. (c) With $df = 1$ and $\alpha = 0.05$ (critical value 3.84), conclude whether Grade Level and Operating System are independent.

(a) Expected Count: Total iOS = 120, Total Android = 80, Grand Total = 200.
$E_{1,1} = \frac{\text{(Row 1 Total)} \times \text{(Col 1 Total)}}{\text{Grand Total}} = \frac{100 \times 120}{200} = \mathbf{60.0}$. All four expected counts are: 9th/iOS: 60, 9th/Android: 40, 12th/iOS: 60, 12th/Android: 40. (All $E \geq 5$). (b) Chi-Square Statistic:
$\chi^2 = \sum \frac{(O-E)^2}{E} = \frac{(70-60)^2}{60} + \frac{(30-40)^2}{40} + \frac{(50-60)^2}{60} + \frac{(50-40)^2}{40}$.

3 **[Foundational]** • **Target: Logistic Regression & Classification Practice Exercises**

Perform a complete bifurcation analysis of the system in terms of parameter $k \in \mathbb{R}$:
$\{x + y + kz = 2; 3x + 4y + 2z = k; 2x + 3y - z = 1$ Determine all values of k for which the system has (a) a unique solution, (b) infinitely many solutions, and (c) no solution.

Step 1: Compute determinant of coefficient matrix A: $\det(A) = |1, 1, k; 3, 4, 2; 2, 3, -1|$. Expanding along row 1: $\det(A) = 1(4(-1) - 2(3)) - 1(3(-1) - 2(2)) + k(3(3) - 4(2)) = 1(-10) - 1(-7) + k(1) = k - 3$. Step 2: Classify cases:(a) Unique Solution: If $keq3$, $\det(A)eq0$; the system has full rank 3 and possesses a unique solution.Critical Case $k = 3$: Substitute $k = 3$ into the augmented matrix.

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4 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Verify the Cayley–Hamilton Theorem for $A = (2, 1; 3, 4)$, and use it to compute A^{-1} without using cofactor formulas or row reduction.

Step 1: Characteristic polynomial: $p(\lambda) = \det(A - \lambda I) = (2 - \lambda)(4 - \lambda) - 3 = \lambda^2 - 6\lambda + 5$. Step 2: Cayley–Hamilton Verification $p(A) = O$: $A^2 = (2, 1; 3, 4)(2, 1; 3, 4) = (4 + 3, 2 + 4; 6 + 12, 3 + 16) = (7, 6; 18, 19)$.

$p(A) = A^2 - 6A + 5I_2 = (7, 6; 18, 19) - (12, 6; 18, 24) + (5, 0; 0, 5) = (7 - 12 + 5, 6 - 6 + 0; 18 - 18 + 0, 19 - 24 + 5) = (0, 0; 0, 0)$. ✓ Step 3: Solve f.

5 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Level 3 · AdvancedLet $h(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$. Prove that h is strictly increasing on $\mathbb{R}$ without using calculus derivatives, deduce that h is injective, and derive the explicit formula for $h^{-1}(x)$ including its domain.

Step 1 · Algebraic Monotonicity: Rewrite $h(x)$ by dividing numerator and denominator by e^x: $h(x) = \frac{1 - e^{-2x}}{1 + e^{-2x}} = \frac{(1 + e^{-2x}) - 2e^{-2x}}{1 + e^{-2x}} = 1 - \frac{2}{e^{2x} + 1}$. As x, $\frac{2}{e^{2x} + 1} \in (0, 2) \Rightarrow h(x) = 1 - \frac{2}{e^{2x} + 1} \in (-1, 1)$. The domain of h^{-1} is $(-1, 1)$. Set $y = \frac{e^{2x} - 1}{e^{2x} + 1}$: $y(e^{2x} + 1) = e^{2x} - 1 \Rightarrow ye^{2x} + y = e^{2x} - 1 \Rightarrow e^{2x}(1 - y) = 1 + y$. For $y \in (-1, 1)$.

6 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Let $A = (1, 2, 0; 0, 3, 1; 2, 1, 4)$. Compute $\det(A)$, find the adjugate matrix $\text{adj}(A)$, and verify the identity $A\text{adj}(A) = \det(A)I_3$.

Step 1: Compute $\det(A)$: Expanding along row 1: $\det(A) = 1(3(4) - 1(1)) - 2(0(4) - 1(2)) + 0 = 1(11) - 2(-2) = 11 + 4 = 15$. Step 2: Compute Cofactor Matrix C: $C_{11} = +(12 - 1) = 11$, $C_{12} = -(0 - 2) = 2$, $C_{13} = +(0 - 6) = -6$ $C_{21} = -(8 - 0) = -8$, $C_{22} = +(4 - 0) = 4$, $C_{23} = -(1 - 4) = 3$ $C_{31} = +(2 - 0) = 2$, $C_{32} = -(1 - 0) = -1$, $C_{33} = +(3 - 0) = 3$ The adjugate matrix is the transpose $\text{adj}(A) = C^T$.

7 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Use the Green's Theorem line integral formula for area, $A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$, to derive the area enclosed by the hypocycloid (astroid) $\mathbf{r}(t) = \langle a\cos^3 t,\ a\sin^3 t\rangle$ for $0 \le t \le 2\pi$.

1. Compute the differentials: $dx = -3a\cos^2 t\sin t\,dt$, $dy = 3a\sin^2 t\cos t\,dt$. 2. Evaluate $x\,dy - y\,dx$: $x\,dy = (a\cos^3 t)(3a\sin^2 t\cos t\,dt) = 3a^2\cos^4 t\sin^2 t\,dt$. $y\,dx = (a\sin^3 t)(-3a\cos^2 t\sin t\,dt) = -3a^2\sin^4 t\cos^2 t\,dt$. $x\,dy - y\,dx = 3a^2\cos^2 t\sin^2 t(\cos^2 t + \sin^2 t)\,dt = 3a^2\cos^2 t\sin^2 t\,dt$. 3. Use trigonometric identities: Since $\cos t\sin t = \frac{1}{2}\sin(2t)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 **[Intermediate]** • **Target: Logistic Regression & Classification Practice Exercises**

Find the surface area of the paraboloid cap $z = x^2 + y^2$ that lies below the horizontal plane $z = 4$.

1. Surface Area Differential for a Graph $z = g(x, y)$: $g_x = 2x, \quad g_y = 2y \Rightarrow dS = \sqrt{1 + (2x)^2 + (2y)^2}\, dA = \sqrt{1 + 4(x^2 + y^2)}\, dA.$ 2. Projection onto xy-plane: The intersection $x^2 + y^2 = 4$ is a disk D of radius 2. In polar coordinates: $0 \le r \le 2$, $0 \le \theta \le 2\pi$, and $dA = r\, dr\, d\theta$. 3. Evaluate the surface integral.

9 **[Intermediate]** • **Target: Logistic Regression & Classification Practice Exercises**

Prove that the limit $\lim_{(x, y) \to (0, 0)} \frac{x^2 y}{x^4 + y^2}$ does not exist by using the Two-Path Test, comparing straight-line approaches with parabolic approaches.

1. Approach along linear paths $y = mx$: $\lim_{x \to 0} \frac{x^2(mx)}{x^4 + (mx)^2} = \lim_{x \to 0} \frac{mx^3}{x^2(x^2 + m^2)} = \lim_{x \to 0} \frac{mx}{x^2 + m^2} = \frac{0}{m^2} = 0 \quad (\text{for } m \neq 0).$ Along $y = 0$ (the x-axis), $\lim_{x \to 0} \frac{0}{x^4} = 0$. Along all straight lines through the origin, the limit is 0. 2. Approach along parabolic paths $y = kx^2$.

10 **[Intermediate]** • **Target: Logistic Regression & Classification Practice Exercises**

In the tomato experiment in Problem 7, the agronomist has two distinct tomato varieties: Roma and Beefsteak. (a) Explain why a randomized block design is superior to a completely randomized design. (b) Outline the blocking protocol.

(a) Superiority of Blocking: Tomato variety is known to have a massive effect on fruit yield. A completely randomized design risks assigning disproportionately more Beefsteak plants to one fertilizer, confounding variety with fertilizer efficacy. Blocking by variety isolates and eliminates the variability between varieties from the experimental error, increasing statistical power. (b) Blocking Protocol: Separate the plants into two blocks: Block 1 (Roma) and Block 2 (Beefsteak). Within Block 1, randomly assign half.

11 **[Intermediate]** • **Target: Logistic Regression & Classification Practice Exercises**

Level 2 · MediumFind the point (x, y) on the parabola $y = x^2$ that is closest to the point $(0, 2)$.

Step 1 · Distance Objective: The distance d from (x, x^2) to $(0, 2)$ is: $d = \sqrt{(x - 0)^2 + (x^2 - 2)^2}$. Minimizing d is equivalent to minimizing the squared distance function $S(x) = d^2$: $S(x) = x^2 + (x^2 - 2)^2 = x^2 + x^4 - 4x^2 + 4 = x^4 - 3x^2 + 4.$ Step 2 · Critical Numbers: $S'(x) = 4x^3 - 6x = 2x(2x^2 - 3) = 0.$ Critical roots: $x = 0$ and $x = \pm\sqrt{\frac{3}{2}} = \pm\frac{\sqrt{6}}{2}$. Step 3 · Second Derivative Test: $S''(x) = 12x^2 - 6.$ • At $x = 0$: $S''(0) = -60$ (certified loc.

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12 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

In an SRS of 200 registered voters, 116 favor a new environmental regulation. (a) Check all three conditions for constructing a 95% confidence interval for the population proportion. (b) Construct the 95% confidence interval. (c) Interpret the interval in context.

(a) Condition Checks:Random: Stated as an SRS of 200 registered voters. (Satisfied).10% Condition: 200 voters is less than 10% of all registered voters ($N \geq 2,000$). (Satisfied).Large Counts: $n\hat{p} = 116 \geq 10$ and $n(1-\hat{p}) = 84 \geq 10$. (Satisfied; normal approximation is valid).(b) 95% Confidence Interval: $\hat{p} = 116/200 = 0.58, z^* = 1.96$. SE $= \sqrt{\frac{0.58(0.42)}{200}} = \sqrt{\frac{0.2436}{200}} \approx 0.0349$. Margin of error: ME $= 1.96(0.0349) \approx 0.0684$. .

13 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

Level 3 · AdvancedEvaluate the definite integral analytically: $\int_{-3}^{3} \left(x^5\sqrt{9-x^2} + \sqrt{9-x^2}\right) dx$.

Split the integral using linearity: $\int_{-3}^{3} x^5\sqrt{9-x^2}\,dx + \int_{-3}^{3} \sqrt{9-x^2}\,dx$. Part 1 · Symmetry Analysis: Let $g(x) = x^5\sqrt{9-x^2}$. Then $g(-x) = (-x)^5\sqrt{9-(-x)^2} = -x^5\sqrt{9-x^2} = -g(x)$. Because g is an odd function integrated over symmetric limits $[-3, 3]$: $\int_{-3}^{3} x^5\sqrt{9-x^2}\,dx = 0$. Part 2 · Geometric Area: The integrand $y = \sqrt{9-x^2} \geq 0$ represents the upper half of the circle $x^2 + y^2 = 9$ with radius $r = 3$. The integral from.

14 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

Level 2 · MediumExplain why the matrix $A = (3, 1; 0, 3)$ is not diagonalizable.

Step 1 · Algebraic Multiplicity: A is upper-triangular, so its characteristic polynomial is $p(\lambda) = (3-\lambda)^2 = 0$. The only eigenvalue is $\lambda = 3$ with algebraic multiplicity $m_a(3) = 2$. Step 2 · Geometric Multiplicity: Compute the eigenspace $E_3 = \text{Null}(A - 3I)$: $A - 3I = (0, 1; 0, 0)$. This matrix has pivot in column 2, with only one free variable $x_1 = t$. The eigenvectors are $\mathbf{v} = t(1; 0)$, so $\dim(E_3) = 1$. Conclusion: Since the geometric mul.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Level 3 · Advanced / University RigorLet A be a 5×8 matrix with $\text{rank}(A) = 4$. (a) What is the dimension of the solution space to $A\mathbf{x} = \mathbf{0}$? (b) Is $A\mathbf{x} = \mathbf{b}$ consistent for every $\mathbf{b} \in \mathbb{R}^5$? Explain. (c) What is the dimension of $\text{Null}(A^T)$?

(a) Null Space Dimension: By the Rank-Nullity Theorem, $\text{rank}(A) + \text{nullity}(A) = n$. Here $n = 8$ and $\text{rank}(A) = 4$: $\text{nullity}(A) = 8 - 4 = \mathbf{4}$. (b) Consistency for all $\mathbf{b}$: No. $\text{Col}(A)$ has dimension 4, which is a proper subspace of $\mathbb{R}^5$ (dimension 5). Therefore, there exist vectors $\mathbf{b} \in \mathbb{R}^5$ that do not lie in $\text{Col}(A)$, making $A\mathbf{x} = \mathbf{b}$ inconsistent. (c) Left Null Space Dimension: The left null space.

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16 **[Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI**

Level 3 · Advanced A tank initially contains 100 liters of pure water. Brine containing 0.5 kg of salt per liter enters the tank at a rate of 4 L/min, and the well-stirred solution leaves the tank at the same rate of 4 L/min. Find the amount of salt $S(t)$ in the tank after t minutes.

Rate of change of salt: $\frac{dS}{dt} = \text{Rate In} - \text{Rate Out}$. Rate In $= (0.5\ \text{kg/L}) \times (4\ \text{L/min}) = 2\ \text{kg/min}$. Since volume remains constant at 100 L, salt concentration is $S(t)/100$.

Rate Out $= \left(\frac{S(t)}{100}\ \text{kg/L}\right) \times (4\ \text{L/min}) = \frac{4S}{100} = 0.04S\ \text{kg/min}$. Set up the linear ODE:

$\frac{dS}{dt} = 2 - 0.04S = -0.04(S - 50)$ Separate variables.

17 **[Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations**

Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be defined by $T(x, y, z) = (x + 2y - z,\ 2x + 5y,\ 3x + 7y - z)$. Find bases for $\ker(T)$ and $\text{im}(T)$, and determine whether T is an isomorphism.

Step 1: Standard matrix A: $A = (1, 2, -1; 2, 5, 0; 3, 7, -1)$. Step 2: Row reduce to RREF: $(1, 2, -1; 2, 5, 0; 3, 7, -1) \to (1, 2, -1; 0, 1, 2; 0, 1, 2) \to (1, 0, -5; 0, 1, 2; 0, 0, 0)$. Step 3: Kernel and Image:$\ker(T)$: Free variable $z = t$. $x = 5t$, $y = -2t$. Basis = $\{(5; -2; 1)\}$, nullity = 1.$\text{im}(T)$: Pivot columns 1 and 2 of A. Basis =.

18 **[Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization**

A coffee brand claims its jars contain a mean of 200 grams of coffee. A consumer agency takes an SRS of 16 jars, finding $\bar{x} = 197.5$ g and $s = 4.0$ g. Assume jar weights are approximately normally distributed. Conduct a significance test of $H_0 : \mu = 200$ vs. $H_a : \mu < 200$ at $\alpha = 0.05$.

1. State: Test $H_0 : \mu = 200$ g vs $H_a : \mu < 200$ g at $\alpha = 0.05$, where μ is the true mean weight of all coffee jars. 2. Plan: 1-sample t-test for mean. Random sample stated; $16 < 10\%$ of all jars; population stated as approximately normal. 3. Do: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{197.5 - 200}{4.0/\sqrt{16}} = \frac{-2.5}{1.0} = -2.50 \quad (df = n - 1 = 15)$. From t-distribution table with $df = 15$: $\text{p} - \text{value} = P(T < -2.50) \approx 0.0123$. 4. Conc.

19 **[Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule**

Eight patients with chronic insomnia try a new relaxation therapy. Their sleep duration (hours/night) is measured before and after 4 weeks of therapy: Patient differences $d = \text{After} - \text{Before}$ yield $\bar{x}_d = 1.25$ hours and $s_d = 0.80$ hours. Assuming normality of differences, test whether the therapy significantly increased sleep at $\alpha = 0.01$.

1. Method Choice: Paired data measured on the same 8 patients $\Rightarrow$ Paired t-test on differences. 2. Hypotheses: $H_0 : \mu_d = 0 \quad \text{vs.} \quad H_a : \mu_d > 0 \quad (\alpha = 0.01)$, where μ_d is the true mean difference in sleep duration (After minus Before). 3. Test Statistic & p-value: $t = \frac{\bar{x}_d - 0}{s_d/\sqrt{n}} = \frac{1.25 - 0}{0.80/\sqrt{8}} = \frac{1.25}{0.2828} \approx 4.42 \quad (df = 8 - 1 = 7)$. With $df = 7$, $P(T > 4.42) \approx 0.0015$. 4. Conclusion: Because.

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20 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Level 3 · Advanced Evaluate the improper integral or establish its divergence: $\int_0^\infty xe^{-x/2}\,dx$

Express as a limit of a definite integral: $\int_0^\infty xe^{-x/2}\,dx = \lim_{t\to\infty}\int_0^t xe^{-x/2}\,dx$ Integrate by parts on $[0, t]$: Let $u = x \Rightarrow du = dx$, and $dv = e^{-x/2}dx \Rightarrow v = -2e^{-x/2}$. $\int_0^t xe^{-x/2}\,dx = [-2xe^{-x/2}]_0^t - \int_0^t(-2e^{-x/2})\,dx = -2te^{-t/2} + 0 + 2[-2e^{-x/2}]_0^t$ $= -2te^{-t/2} - 4(e^{-t/2} - 1) = 4 - e^{-t/2}(2t + 4)$ Take the limit as $t \to \infty$: By L'Hôpital's Rule.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Logistic model: $\frac{dP}{dt} = rP(1 - P/K)$. At $P = K/2$, the growth rate is:

a. zero

b. minimum

c. **maximum** ✓

d. negative

Applying Differential Equations principles, Maximum growth at $P = K/2$. Therefore, the verified correct option is maximum.

22 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Carrying capacity K in the logistic model: as $P \to K$ from below, growth rate $dP/dt \to$

a. 0 ✓

b. K

c. ∞

d. -1

Applying Differential Equations principles, At $P = K$: $dP/dt = rK(1 - K/K) = 0$. Therefore, the verified correct option is 0.

23 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

In a bivariate regression with sample size $n = 31$ (Research Model 1), sample slope is $b_1 = 1.70$ with $\text{SE}_{b_1} = 0.26$. Testing $H_0 : \beta_1 = 0$, what is the t-test statistic and degrees of freedom?

a. $t = 6.54$, $\text{df} = 29$ ✓

b. $t = 0.44$, $\text{df} = 29$

c. $t = 8.08$, $\text{df} = 31$

d. $t = 6.54$, $\text{df} = 30$

$t = \frac{b_1 - 0}{\text{SE}_{b_1}} = \frac{1.70}{0.26} = 6.54$, with $\text{df} = n - 2 = 31 - 2 = 29$.

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24 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

In a regression analysis with $n = 23$ data points, the slope estimate is $b_1 = 2.31$ with standard error $SE_{b_1} = 0.909$. Find the test statistic t for testing $H_0 : \beta_1 = 0$.

a. 1.04

b. 4.54

c. 1.54

d. **2.54** ✓

$t = \frac{b_1}{SE_{b_1}} = \frac{2.31}{0.909} = 2.54$ with $df = n - 2 = 21$.

25 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

A regression line predicts $\hat{y} = 41.85$ for $x = 25$, while the observed value is $y = 39.1$. What is the residual for this point?

a. 3.25

b. **-2.75** ✓

c. -0.75

d. -6.75

Residual $= y - \hat{y} = 39.1 - 41.85 = -2.75$.

26 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Inflection point classification: $f''(c) = 0$ is necessary but not sufficient. Example of $f''(0) = 0$ with no inflection:

a. x^4 ✓

b. x^3

c. x^2

d. $\sin x$

x^4: $f''(0) = 0$ but concave up everywhere.

27 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Logistic equation $P' = rP(1 - P/K)$ has equilibria at:

a. $P = 0$ only

b. $P = K$ only

c. $P = r$

d. $P = 0$ **and** $P = K$ ✓

Equilibria: $P' = 0 \Rightarrow P = 0$ or $P = K$.

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28 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Logistic equation steady states (equilibria):

a. $P = 0$ only

b. $P = K$ only

c. $P = 0$ **and** $P = K$ ✓

d. $P = K/2$

$rP(1 - P/K) = 0 \Rightarrow P = 0$ or $P = K$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

The least-squares regression line for a data set is $\hat{y} = 11.1 + 2.5x$. What is the predicted value of y when $x = 45$?

a. 128.6

b. **123.6** ✓

c. 118.6

d. 125.6

$\hat{y} = 11.1 + 2.5(45) = 123.6$.

30 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

Evaluate the double integral $\iint_R xy\,dA$ where R is the region in the first quadrant bounded by the lines $y = x$, $x = 1$, and $y = 0$.

a. 1/12

b. **1/8** ✓

c. 1/6

d. 1/4

The region of integration R is a triangle with vertices (0,0), (1,0), and (1,1). We can set up the double integral using Fubini's Theorem. Since $y = x$ is the upper bound for y and $x = 1$ is the upper bound for x, we can integrate with respect to y first, then x. The limits for y are from 0 to x, and the limits for x are from 0 to 1. $\iint_R xy\,dA = \int_0^1 \int_0^x xy\,dy\,dx$ First, evaluate the inner integral with respect to y.

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31 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

The histogram below displays the distribution of the number of hours students spent studying for a recent exam. What is the approximate median number of hours studied?

a. 2.5 hours

b. **3.5 hours** ✓

c. 4.5 hours

d. 5.5 hours

The histogram shows the frequency of students studying within certain hour intervals. To find the median, we need to locate the value that divides the data into two equal halves. First, calculate the total number of students represented by the histogram. The heights of the bars represent frequencies. Let's assume the heights are as follows (this would be explicit in a real AP question with a diagram): Interval [0, 1) has frequency 5, [1, 2) has frequency 10, [2, 3) has frequency 18, [3, 4) has frequency 22, [4, 5).

32 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

Consider the region R in the xy-plane bounded by the curves $y = x^2$ and $y = 4$. What is the area of this region?

a. $\frac{32}{3}$ ✓

b. $\frac{16}{3}$

c. 8

d. 16

The area of the region bounded by two curves $y = f(x)$ and $y = g(x)$ from $x = a$ to $x = b$, where $f(x) \geq g(x)$ on $[a, b]$, is given by the integral $\int_a^b (f(x) - g(x))dx$. First, we find the intersection points of $y = x^2$ and $y = 4$ by setting them equal: $x^2 = 4$, which gives $x = -2$ and $x = 2$. So, our limits of integration are $a = -2$ and $b = 2$. On the interval $[-2, 2]$, the line $y = 4$ is above the parabola $y = x^2$. Therefore, the area A is given by: $A = \int_{-2}^{2} (4 - x^2)dx$ Now, we eva.

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33 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

The figure shows the graph of a continuous function $f(x)$ on the interval $[a, d]$. The shaded region represents the definite integral $\int_a^c f(x)\,dx$. If the area of the shaded region is 12 square units and the area of the unshaded region between $x = c$ and $x = d$ is 5 square units, what is the value of $\int_a^d f(x)\,dx$?

a. **17** ✓

b. 7

c. 2.4

d. -7

The Fundamental Theorem of Calculus relates the definite integral of a function to its antiderivative. Geometrically, the definite integral $\int_a^b f(x)\,dx$ represents the net signed area between the curve $f(x)$ and the x-axis from $x = a$ to $x = b$. Areas above the x-axis are positive, and areas below the x-axis are negative. The problem states that the area of the shaded region, representing $\int_a^c f(x)\,dx$, is 12 square units. This implies that this region contributes positively to the net signe.

34 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

Consider the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$. If the radius of convergence is $R = 3$ and the center of the series is $a = 2$, which of the following intervals represents the values of x for which the series converges?

a. $(-1, 5)$ ✓

b. $[-1, 5]$

c. $(-1, 5]$

d. $[-1, 5)$

The radius of convergence R of a power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ indicates that the series converges absolutely for $|x-a|R$. The center of the series is a. In this problem, we are given $R = 3$ and $a = 2$. Thus, the series converges for $|x-2| < 3$. This inequality can be rewritten as $-3 < x-2 < 3$. Adding 2 to all parts of the inequality, we get $-3+2 < x < 3+2$, which simplifies to $-1 < x < 5$. This interval is written in open interval notation as $(-1, 5)$. The convergence at the endpoints.

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35 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

Consider the region R in the first quadrant bounded by the x-axis, the y-axis, and the line $y = -2x + 6$. We are interested in evaluating the double integral $\iint_R f(x, y)\, dA$ where $f(x, y) = x + y$. However, the integral is set up in polar coordinates as $\int_{\alpha}^{\beta} \int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta) \cdot r\, dr\, d\theta$. If this integral is evaluated from $\theta = 0$ to $\theta = \frac{\pi}{2}$, and the inner integral bounds are $r_1(\theta) = 0$ and $r_2(\theta) = \frac{6}{\sin\theta + 2\cos\theta}$, and the outer integral is $\int_0^{\pi/2} \int_0^{\frac{6}{\sin\theta + 2\cos\theta}} (r\cos\theta + r\sin\theta) \cdot r\, dr\, d\theta$, determine the value of α and β that correspond to the description of region R in polar coordinates for $0 \le \theta \le \frac{\pi}{2}$.

a. $\alpha = 0, \beta = \frac{\pi}{2}$ ✓

b. $\alpha = 0, \beta = \frac{\pi}{4}$

c. $\alpha = 0, \beta = \frac{3\pi}{4}$

d. $\alpha = \frac{\pi}{4}, \beta = \frac{\pi}{2}$

The region R is in the first quadrant, bounded by the x-axis ($y = 0$), the y-axis ($x = 0$), and the line $y = -2x + 6$. When $x = 0$, $y = 6$. When $y = 0$, $2x = 6$, so $x = 3$. The vertices of the region are $(0, 0), (3, 0), (0, 6)$. In polar coordinates, $x = r\cos\theta$ and $y = r\sin\theta$. The line $y = -2x + 6$ becomes $r\sin\theta = -2r\cos\theta + 6$. Rearranging for r, we get $r(\sin\theta + 2\cos\theta) = 6$, so $r = \frac{6}{\sin\theta + 2\cos\theta}$. The first quadrant is defined by.

36 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

Consider the slope field shown below. Which of the following differential equations could represent this slope field, assuming the slopes are defined at all points except where division by zero occurs? $\frac{dy}{dx} = \frac{x}{y}$

a. $\frac{dy}{dx} = \frac{x}{y}$ ✓

b. $\frac{dy}{dx} = \frac{y}{x}$

c. $\frac{dy}{dx} = x - y$

d. $\frac{dy}{dx} = x + y$

The slope field shows that at points where $y = 0$ (the x-axis), the slopes are undefined (vertical lines). This is consistent with the differential equation $\frac{dy}{dx} = \frac{x}{y}$, as division by zero occurs when $y = 0$. Additionally, observe the slopes in different quadrants. In the first quadrant ($x > 0, y > 0$), the slopes are positive, consistent with $\frac{x}{y} > 0$. In the second quadrant ($x0$), the slopes are negative, consistent with $\frac{x}{y}0$. In the fourth quadrant ($x > 0, y < 0$), the slopes are.

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37 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

Consider the region R in the first quadrant bounded by the x-axis, the line $y = x$, and the circle $x^2 + y^2 = 16$. The integral $\iint_R f(x, y)\, dA$ can be evaluated more easily in polar coordinates. If $f(x, y) = \frac{1}{\sqrt{x^2 + y^2}}$, and the integral in polar coordinates is given by $\int_0^{\pi/4} \int_0^4 r \cdot \frac{1}{r}\, dr\, d\theta = \int_0^{\pi/4} A\, d\theta$, find the value of A.

a. **4** ✓

b. $\pi/4$

c. 16

d. 2

The region R is described in the first quadrant, bounded by the x-axis ($\theta = 0$), the line $y = x$ ($\theta = \pi/4$), and the circle $x^2 + y^2 = 16$ ($r = 4$). The integrand is $f(x, y) = \frac{1}{\sqrt{x^2 + y^2}}$. In polar coordinates, $x^2 + y^2 = r^2$, so $\sqrt{x^2 + y^2} = r$. Thus, $f(x, y) = \frac{1}{r}$. The differential area element in polar coordinates is $dA = r\, dr\, d\theta$. The double integral in polar coordinates is.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

The figure shows the graph of the function $f(x) = x^2 - 2x - 1$ and the point $P(3, 2)$ on the curve. What is the slope of the tangent line to the graph of f at point P?

a. **4** ✓

b. 2

c. 3

d. 6

The slope of the tangent line to the graph of $f(x)$ at any point is given by its derivative, $f'(x)$. 1. Find the derivative of $f(x) = x^2 - 2x - 1$ using the power rule: $f'(x) = 2x - 2$ 2. Evaluate the derivative at the x-coordinate of point $P(3, 2)$, which is $x = 3$: $f'(3) = 2(3) - 2 = 6 - 2 = 4$ Thus, the slope of the tangent line at P is 4. Misconception Analysis: - 2 results from mistakenly reading the y-coordinate of point P as the slope, since $f(3) = 2$. - 3 results from taking the x-c.

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39 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Consider the region bounded by the curves $y = x^2$ and $y = 4$. This region is revolved around the line $y = 5$ to form a solid. A cylindrical hole is drilled through the center of this solid, perpendicular to the axis of revolution. The radius of the cylindrical hole is r. The volume of the material removed by the cylindrical hole is given by $V_{\text{hole}} = \pi\int_a^b (R(x)^2 - r^2)dx$, where $R(x)$ is the outer radius of the solid at a given x and a, b are the limits of integration. Find the value of a and b, and the expression for V_{hole} in terms of r. The question asks for the setup of the integral representing the volume of the material removed.

a. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((5 - x^2)^2 - r^2)dx$ ✓

b. Limits of integration are 0 and 4; $V_{\text{hole}} = \pi\int_0^4((4 - x^2)^2 - r^2)dx$

c. Limits of integration are 1 and 4; $V_{\text{hole}} = \pi\int_1^4((5 - x^2)^2 - r^2)dx$

d. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((4 - x^2)^2 - r^2)dx$

The region is bounded by $y = x^2$ and $y = 4$. To find the intersection points, we set $x^2 = 4$, which gives $x = \pm 2$. These are our limits of integration, $a = -2$ and $b = 2$. The region is revolved around the line $y = 5$. The outer radius $R(x)$ is the distance from the axis of revolution ($y = 5$) to the outer boundary of the region. The outer boundary is the curve $y = x^2$. So, $R(x) = 5 - x^2$. The inner radius of the cylindrical hole is given as r. The volume of the material removed by the cylindrical.

40 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

The graph of the quadratic function $f(x) = x^2 - 6x + k$ is shown. The vertex of the parabola is at the point $(3, -5)$. Find the value of the parameter k.

a. -5

b. 4 ✓

c. 9

d. 0

The vertex of a parabola in the form $f(x) = ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$. For the given function $f(x) = x^2 - 6x + k$, we have $a = 1$ and $b = -6$. Thus, the x-coordinate of the vertex is $x = -\frac{-6}{2(1)} = 3$. This matches the given vertex $(3, -5)$. Since the vertex $(3, -5)$ lies on the graph of the function, it must satisfy the function's equation. Substitute $x = 3$ and $f(x) = -5$ into the equation: $-5 = (3)^2 - 6(3) + k$ $-5 = 9 - 18 + k$ $-5 = -9 + k$ To solve for k, add 9 to both sides.

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41 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

The graph of a differentiable function $f(x)$ is shown below. The function has a tangent line at $x = 1$ with equation $y = 2x + c$. If $f(1) = 3$, find the value of c.

a. **1** ✓

b. 2

c. 3

d. 4

The equation of the tangent line to $f(x)$ at $x = 1$ is given as $y = 2x + c$. We are given that $f(1) = 3$. This means the point $(1, 3)$ lies on both the function $f(x)$ and its tangent line at $x = 1$. Since the point $(1, 3)$ lies on the tangent line $y = 2x + c$, we can substitute the coordinates into the equation: $3 = 2(1) + c$ $3 = 2 + c$ $c = 3 - 2$ $c = 1$ The derivative of the function at $x = 1$ is the slope of the tangent line, which is 2. This information is consistent with the problem but not direct.

42 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Consider the graph of the function $f(x)$ shown below. Which of the following points lies on the graph of $f(x)$?

a. $(-2, 3)$

b. $(0, -1)$

c. $(1, 0)$ ✓

d. $(3, 4)$

The graph of $f(x)$ passes through the point $(1, 0)$. To verify this, we can observe the graph and identify the coordinates where the curve intersects the x-axis at $x = 1$. Let's check the other options. The point $(-2, 3)$ is not on the curve as the function value at $x = -2$ is approximately 5. The point $(0, -1)$ is not on the curve as the function value at $x = 0$ is approximately 3. The point $(3, 4)$ is not on the curve as the function value at $x = 3$ is approximately 0. Therefore, only $(1, 0)$ lies on the.

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43 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

The figure shows the region R bounded by the curves $y = x^2$ and $y = 4$. Which of the following double integrals represents the area of region R using the order of integration $dydx$?

a. $\int_{-2}^{2}\int_{x^2}^{4} dydx$ ✓

b. $\int_{-2}^{2}\int_{4}^{x^2} dydx$

c. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dydx$

d. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dxdy$

The region R is bounded above by $y = 4$ and below by $y = x^2$. To find the limits for x, we find the intersection points of the two curves: $x^2 = 4 \Rightarrow x = \pm 2$. Thus, the region is defined for x from -2 to 2. When integrating with respect to y first ($dydx$), the inner integral goes from the lower curve to the upper curve, so from $y = x^2$ to $y = 4$. The outer integral is over the range of x, which is -2 to 2. Therefore, the double integral representing the area is.

44 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

The region bounded by the curve $y = x^2 - 4$ and the x-axis is rotated about the x-axis. If the volume of the resulting solid is $V = \pi\int_{a}^{b}(x^2 - 4)^2 dx$, find the value of $b - a$.

a. 2

b. **4** ✓

c. 8

d. 16

The region is bounded by the curve $y = x^2 - 4$ and the x-axis. To find the limits of integration, we need to find the x-intercepts of the curve, where $y = 0$. Setting $x^2 - 4 = 0$, we get $x^2 = 4$, which gives $x = \pm 2$. Therefore, the bounds of integration are $a = -2$ and $b = 2$. The volume of the solid of revolution is given by $V = \pi\int_{a}^{b}[f(x)]^2 dx$. In this case, $f(x) = x^2 - 4$, and the limits are $a = -2$ and $b = 2$. The question asks for the value of $b - a$. $b - a = 2 - (-2) = 2 + 2 = 4$.

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45 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Consider the function $f(x) = x^2 - 6x + 5$. The function has a local minimum at which value of x?

a. 0

b. **3 ✓**

c. 6

d. 5

To find the local minimum of the function $f(x) = x^2 - 6x + 5$, we first find the derivative of the function with respect to x: $f'(x) = 2x - 6$. Next, we set the derivative equal to zero to find the critical points: $2x - 6 = 0$. Solving for x, we get $2x = 6$, which means $x = 3$. To determine if this critical point is a local minimum, we can use the second derivative test. The second derivative of $f(x)$ is $f''(x) = 2$. Since $f''(3) = 2 > 0$, the function has a local minimum at $x = 3$.

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